

Roadmap to Inclusive Growth



For Prelims: **Viksit Bharat-Guarantee for Rozgar and Ajeevika Mission (Gramin) Act, 2025** , **Labour Codes** , **World Bank** , **Gini Index**, **Household Consumer Expenditure Survey (HCES 2023-24)** , **Pradhan Mantri Garib Kalyan Yojana (PMGKY)**, **GST** , **Demographic Dividend** , **NITI Aayog** , **Data Centers** , **AI** , **BharatNet** , **Regional Rural Banks (RRBs)** .

For Mains: Key facts regarding inequality and inequality trends in India, Causes and consequences of inequality in India and steps needed to ensure sustainable and inclusive growth.

Source: TH

Why in News?

Recent shifts in India's socio-economic policy, including the introduction of the **Viksit Bharat-Guarantee for Rozgar and Ajeevika Mission (Gramin) Act, 2025** (VB G RAM G Act, 2025) and the **new Labour Codes** , have intensified the debate surrounding rural welfare and the widening **inequality** .

Summary

- India's evolving socio-economic landscape is marked by a declining Gini Index but persistent **urban-rural and class-based disparities** .
- The newly introduced **VB G RAM G Act, 2025**, aims to modernize rural welfare by focusing on high-tech skills and climate resilience.
- To ensure **inclusive growth** , India must address wealth concentration and bridge the "skills gap" to utilize its demographic dividend effectively.

What are the Key Facts Regarding Inequality Trends in India?

- Inequality Underestimation:** While the **World Bank** cites a **Gini index** of **0.25** for India, the **NSSO's Household Consumer Expenditure Survey (HCES 2023-24)** suggests a higher **Gini index of 0.29** ,

indicating that actual consumption inequality is more pronounced than commonly reported.

- **Urban vs. Rural Disparity:** Inequality is significantly higher in **urban sectors** compared to rural ones. The mean **Monthly Per Capita Expenditure (MPCE)** of the **top 10%** in urban areas is **9 times** that of the **bottom 10%** in rural areas.
- **Non-Food Expenditure Gap:** India's consumption boom is driven by **non-food expenditure** , where the disparity is most striking. The urban top 10% accounts for **27% of total non-food spending** , leaving the remaining 90% with a disproportionately smaller share.
- **Class-Based Divergence:** Since the 1980s, **urban owners, managers, and professionals** have gained disproportionately, while **urban informal workers, small farmers, and agricultural labourers** have faced stagnation, leading to a surge in **between-class inequality** .
- **Social Safety Net Overlap:** Data reveals a mismatch in welfare targeting, showing that **one-fourth of the richest 10%** in India benefited from the **Pradhan Mantri Garib Kalyan Yojana (PMGKY)** and 13% of them still hold **BPL ration cards** .
- **Debt-Led Consumption:** A significant portion of Indian households continue to sustain their standard of living through **debt-led consumption** , highlighting a fragile financial foundation despite the "consumption boom."

Inequality

- **About:** Inequality refers to the **unequal distribution** of resources, opportunities, rights, and outcomes among individuals or groups in a society. It is a situation where people do not have equal access to **wealth, education, healthcare, power, justice** , or **social status** .
- **Main Types of Inequality:**

Type	Meaning	Examples
Economic Inequality	Unequal distribution of income and wealth	The top 10% of earners capture 58% of national income, while the bottom 50% receive only 15% . Wealth concentration is higher: the richest 1% hold 40% of total wealth.
Social Inequality	Differences based on caste, class, religion, gender, ethnicity	46% of rural households are landless; landlessness is significantly higher in villages with larger Scheduled Caste (SC) and Scheduled Tribe (ST) populations.
Political Inequality	Unequal access to political power and participation	Marginalized groups having less political representation. E.g., Women 's representation in the 18 th Lok Sabha (2024-2029) stands at approximately at 14% only .
Regional Inequality	Disparity between different regions of a country	Maharashtra leads with a GDP of Rs 24.11 Lakh Cr , while Bihar remains at the bottom with the lowest per capita income (Rs 66,828 in 2023-24) and high multidimensional poverty (33.76%).
Gender Inequality	Unequal treatment and opportunities between men and women	Female Labour Force Participation remains low at ~32-35% . Women globally earn only 61% of what men earn per working hour.

- **Gini Index:** Developed in **1912 by Corrado Gini** , the index is a standard measure of **statistical dispersion** used to represent the income or wealth distribution of a nation's residents. It serves as a vital statistical barometer for **income inequality** .
 - **Lorenz Curve Connection:** The index is mathematically derived from the **Lorenz Curve** , which plots cumulative income against cumulative recipients; the Gini coefficient measures the area between this curve and the **45-degree line of perfect equality** .
 - **Numerical Scale:** Values range from **0 to 1** (or 0 to 100%), where **0** represents **perfect equality** and **1 (or 100)** represents **absolute inequality** (one person holds all the wealth).
 - **India's Declining Trend:** India has shown consistent progress, with its Gini Index improving from **28.8 in 2011 to 25.5 in 2022** , placing the country in the "**moderately low inequality**" bracket. At a score of **25.5** , India demonstrates lower income inequality than major global economies, including the **United States (41.8)** and **China (35.7)** .
- **Inclusive Growth:** Inclusive growth is economic growth that is **distributed fairly across society** and creates opportunities for all, ensuring that the benefits of a growing economy reach every section of the population, particularly the marginalized.
 - This growth is not only rapid but also b **road-based, participatory, and sustainable** , leading to a reduction in poverty and inequality.
 - Key features of inclusive growth include **equitable distribution, employment generation, poverty reduction, social inclusion** , and **regional balance** .

What are the Causes and Consequences of Inequality in India?

Category	Causes	Consequences
Economic	Jobless Growth: Rapid growth in capital-intensive sectors (like IT/Finance) compared to labor-intensive ones (Agriculture/Manufacturing).	Wealth Concentration: The top 1% now holds 40% of India's total wealth , leading to the "Billionaire Raj" phenomenon.
	Regressive Taxation: Heavy reliance on indirect taxes (like GST), which affect the poor more than the rich.	Poverty Traps: Low-income families remain stuck in a cycle of debt and lack of capital to start businesses.
Social & Structural	Caste System: Historical stratification limits social mobility; Scheduled Castes (SCs) and Tribes (STs) face higher rates of landlessness.	Social Fragmentation: Increased polarization and "ghettoization" where different classes and castes live in separate physical and social spaces.
	Land Inequality: Approximately 46% of rural households are landless , while the top 10% own 44% of rural land.	Agrarian Distress: Small farmers lack collateral for credit, leading to high suicide rates and rural-to-urban migration.
Human Development	Unequal Access to Education: Quality private education is expensive , while public schools often lack resources.	Skills Gap: A "lost generation" of youth who are educated but unemployable, hindering India's demographic dividend .
	High Out-of-Pocket Health Costs: Lack of universal healthcare; health emergencies push 63 million Indians into poverty annually.	Reduced Life Expectancy: Poorer deciles suffer from higher morbidity and lower life expectancy compared to the affluent.
Gender	Low Female Labor Force Participation: Stagnant at 32-35% , driven by safety concerns and the burden of unpaid care work.	Economic Loss: Massive loss in potential GDP; women earn only 32% of what men earn when unpaid domestic work is included.
Environmental	Climate Injustice: High consumption by the rich drives carbon emissions (top 1% cause 41% of emissions).	Vulnerability: The world's poorest emit the least yet suffer the most, losing livelihoods to extreme weather and becoming forced climate migrants .

What Steps are Needed to Curb Inequality and Ensure Inclusive Growth?

- **Closing the Digital and Skill Divide:** Pivot from general literacy to "**bridge programs**" that transition informal workers and rural youth into the **AI and green-tech workforce** . Fully implement **BharatNet** to ensure a village health center or school has the same **high-speed infrastructure** as an urban hub, preventing a "digital caste system."
- **Structural Support for Gender Parity:** Mandate universal access to **creches and safe transit** to decouple a woman's career potential from her domestic or geographical constraints. Incentivize **flexible work models** specifically designed to bring the millions of women currently in unpaid domestic work into the **formal, paid economy** .
- **Climate Justice & Rural Resilience:** Use the **VB GRAM G Act, 2025** to fund the creation of community-owned, **climate-resilient assets** (like solar pumps or water conservation tanks) for small-scale farmers. Prioritize **price discovery and credit** for marginal farmers, preventing large-scale agribusiness from monopolizing market gains.
- **Inclusive Circularity:** Formalize the informal waste-picker sector into the new "**resource recovery**" industrial strategy, turning a marginalized labor force into protected, **green-collar workers** .
- **Financial Equity:** Use the **Aadhaar-UPI-DigiLocker** ecosystem to provide low-interest credit and micro-insurance, breaking the cycle of high-interest **predatory lending** in poor districts.
- **Empowering Local Bodies:** Shift financial power from central hubs to local panchayats and **Regional Rural Banks (RRBs)** , allowing for hyper-local solutions to poverty that a "one-size-fits-all" federal policy might miss.

Conclusion

While India's declining Gini Index suggests improving **social equity, deep-seated** structural issues like the **urban-rural consumption gap** and regressive taxation persist. Achieving **Viksit Bharat @ 2047** necessitates transitioning from "**jobless growth**" to a human-centric model that prioritizes high-tech skilling, women-led development, and climate-resilient rural livelihoods.

Drishti Mains Question:

Discuss the significance of the Lorenz Curve and Gini Index in measuring economic inequality. Does India's current score accurately reflect its ground realities?

Frequently Asked Questions (FAQs)

1. What is the Gini Index and the Lorenz Curve?

The Lorenz Curve is a graphical representation of wealth distribution, while the Gini Index is the mathematical coefficient derived from the area between that curve and the line of perfect equality.

2. Why is "Jobless Growth" a major cause of inequality in India?

It occurs when the economy grows through capital-intensive sectors (like IT and Finance) that employ fewer people, rather than labor-intensive sectors (like Manufacturing) that can absorb the mass workforce.

3. What is "Climate Injustice" in the context of economic inequality?

It refers to a situation where the richest 1% drive the highest carbon emissions through consumption, while the poor suffer the most from climate-induced disasters like floods and heatwaves.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/OHN2IW8DMJU](https://www.youtube.com/embed/OHN2IW8DMJU)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Inclusive growth as enunciated in the Eleventh Five Year Plan does not include one of the following: (2010)

- (a) Reduction of poverty
- (b) Extension of employment opportunities
- (c) Strengthening of capital market
- (d) Reduction of gender inequality

Ans: (c)

Mains

Q. COVID-19 pandemic accelerated class inequalities and poverty in India. Comment. (2020)

India's 1st Orbital Data Centre Satellite Pathfinder



Why in News?

Pixxel, a Bengaluru-based satellite imaging startup, has partnered with **Indian AI firm Sarvam AI** to develop **India's first "orbital data centre" satellite named *Pathfinder***. Scheduled for launch by the fourth quarter of 2026, the mission aims to place data-centre-grade **Graphics Processing Units (GPUs)** in **low Earth orbit** to process artificial intelligence workloads directly in space.

What is an Orbital Data Centre?

- **About:** An orbital data centre is a **network or constellation of satellites** equipped with powerful computing hardware (like GPUs) designed to process data directly in space.
 - Traditional satellites generally collect raw data (images, signals) and transmit massive volumes of it back to Earth for analysis. Orbital data centres employ **edge computing**, processing the **data at the source and sending the actionable insights back to ground stations**.
- **Significance:**
 - **Terrestrial Constraints:** Ground-based data centres are increasingly hitting limits regarding **land availability, water for cooling, and energy consumption**, all of which are exacerbated by the massive power demands of **generative AI**.
 - **Continuous Energy:** In the right orbit, **solar power is effectively continuous**.
 - This offers **"free" electricity**, which is the strongest economic argument for moving high-intensity computation to space.
 - **Data Bottlenecks:** Earth observation satellites generate massive, high-resolution files that are expensive and slow to downlink.
 - Processing this data in orbit eases the bandwidth bottleneck.
 - **Strategic Scaling:** Future iterations of **Starlink (V3)** combined with the high-payload capacity of Starship could potentially deliver up to 100 GW of power to orbit, making large-scale space-based computing constellations commercially viable.

Challenges of Computing in Space

- **Thermal Management (Heat Dissipation):** Space is a vacuum, preventing **convection-based cooling** (the natural process of heat transfer on Earth).
 - Heat generated by GPUs must therefore be dissipated through radiative cooling systems using heat-transfer loops and radiating panels.
- **Radiation Exposure:** Commercial GPUs are highly susceptible to **cosmic radiation**, which can cause **"bit flips"** (random changes in computer code) and long-term semiconductor degradation. Shielding these components adds weight and complexity.
- **Miniaturisation Constraints:** Fitting power-hungry data centre components into a small 200-kg satellite requires extraordinary engineering in payload miniaturisation and power management.
- **High Initial Costs:** Launching and maintaining advanced computing hardware in orbit involves significant capital expenditure compared to terrestrial cloud servers.
 - Unlike a terrestrial centre, a satellite cannot be easily repaired. High levels of redundancy must be

built in, and the system must account for "**eclipse periods**" where solar power is unavailable.

- **Scalability:** According to Pixxel, nearly 100-500 satellites may eventually be required to match the computing capability of a terrestrial data centre.
 - While edge processing in space is considered feasible in the near term, large-scale replacement of Earth-based cloud infrastructure remains a long-term possibility.

What is the Pathfinder Mission?

- **About:** Pathfinder is a **200-kg class satellite** designed to function as a demonstrator for in-orbit data processing.
- **Key Payloads & Technology:**
 - **Datacentre-grade GPUs:** Pathfinder will carry high-performance graphics processing units (GPUs) similar to those used in terrestrial data centres.
 - **Hyperspectral Imaging:** It will carry Pixxel's flagship **hyperspectral imaging camera**, capable of capturing high-fidelity data of the Earth.
 - **Sovereign AI Backbone:** Sarvam AI will provide **full-stack language models** and an AI inference platform. This allows AI training and inference to occur directly on the satellite.
- **Objective:** To validate real-time AI inference, power management, and thermal constraints in the harsh space environment, establishing the technical groundwork for commercial-scale orbital data centres.
- **Manufacturing:** The satellite will be built at **Gigapixxel**, Pixxel's new manufacturing facility designed to scale satellite production.

Note: Apart from this Indian space mission, the name "**Pathfinder**" is also famously associated with **NASA's Mars Pathfinder (1997)**, a historic mission that successfully landed the first robotic rover, Sojourner, on the surface of Mars.

Frequently Asked Questions (FAQs)

1. What is an orbital data centre?

An orbital data centre is a satellite or satellite network equipped with powerful GPUs and computing systems to process data directly in space. It reduces dependence on ground stations by sending only useful insights to Earth.

2. Why is India's Pathfinder mission important?

Pathfinder is India's first orbital data centre satellite and a major step in combining AI, hyperspectral imaging, and space computing. It will test whether advanced workloads can be run reliably in low Earth orbit.

3. What role does Sarvam AI play in the mission?

Sarvam AI will provide the AI backbone, including full-stack language models and inference platforms. This will help run AI training and processing directly on the satellite.

4. What are the biggest challenges in space-based computing?

The main challenges are heat dissipation, radiation damage, limited maintenance, power storage during eclipse periods, and the high cost of launching and operating hardware in orbit.

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following? (2020)

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

- (a) 1, 2, 3 and 5 only
- (b) 1, 3 and 4 only
- (c) 2, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

Ans: (b)

Q. Satellites used for telecommunication relay are kept in a geostationary orbit. A satellite is said to be in such an orbit when: (2011)

1. The orbit is geosynchronous.
2. The orbit is circular.
3. The orbit lies in the plane of the Earth's equator.
4. The orbit is at an altitude of 22,236 km.

Select the correct answer using the codes given below:

- a) 1,2 and 3 only
- b) 1,3 and 4 only
- c) 2 and 4 Only
- d) 1, 2, 3 and 4

Ans: (a)

India-Algeria Defence Cooperation



Source: TH

Recently, **India and Algeria** held the **inaugural Joint Commission on Defence Cooperation meeting** in New Delhi, reflecting growing bilateral defence ties following the signing of the **India-Algeria Defence MoU in 2024**.

- Discussions focused on **military training, joint exercises, defence industry collaboration and medical cooperation**.
- Both sides signed the **Rules of Procedure** to streamline implementation and future engagement under the India-Algeria defence cooperation framework.

Algeria

- Algeria is located in **North Africa**, bordered by the **Mediterranean Sea**, **Morocco**, **Tunisia**, **Libya**, **Mali**, **Niger** and **Mauritania**.
- It is the **largest country in Africa by area** and includes a major portion of the **Sahara Desert**.
- Roughly 80-90% of Algeria's territory lies within the Sahara, giving it **one of the largest desert areas** of any country in the world.
- This vast desert landscape strongly shapes Algeria's climate and underpins its **hydrocarbon - based economy**, particularly oil and natural gas production, as well as distinctive desert - adapted cultural practices.



Read more: [India-Algeria Strengthen Defence Ties](#)

SC Empowers District Collectors to Enforce SWM Rules, 2026



Source:TH

The **Supreme Court of India** directed the **Ministry of Environment, Forest and Climate Change (MoEFCC)** to delegate specific statutory powers to **District Collectors** under the **Environment (Protection) Act, 1986** , for strict implementation of the **Solid Waste Management (SWM) Rules, 2026** , emphasising the need to protect the planet from “man-made destruction”.

- The directions came during a hearing related to the **Bhopal Municipal Corporation** , which was found violating the earlier **SWM Rules, 2016**, by the **National Green Tribunal (NGT)**.
- **Delegation of Powers:** District Collectors across the country were instructed to establish special monitoring cells comprising relevant officials for a **period of one year** to ensure effective implementation of waste management measures.
 - They were also empowered to conduct virtual **inspections** of dumping sites and disconnect water and **electricity supply to bulk waste generators** violating the rules.
- **Compliance and Monitoring Mechanism:** The Court directed District Collectors to prepare **fortnightly compliance reports** and submit them to designated State authorities, ensuring continuous monitoring and administrative accountability in waste management implementation.
- **Directions to State Governments:** State governments were directed to incentivise **well-performing urban and rural local bodie s** through grants, while defaulting local authorities would face penalties for non-compliance with the SWM Rules, 2026.
- **Tourist and Pilgrimage Waste Management:** The Court asked Chief Secretaries to formulate dedicated waste management mechanisms for tourist destinations and pilgrimage centres to improve sanitation standards and strengthen India’s global image.
- **Institutional and Financial Challenges:** The Court acknowledged challenges such as budgetary shortages, manpower constraints, and weak institutional capacities at the grassroots level.
 - It also suggested creating a **dedicated cadre within rural local bodies** for strengthening waste management systems.
- **CSR and Waste-to-Energy Initiatives:** Authorities were directed to explore the use of **Corporate Social Responsibility (CSR)** funds for establishing **compressed biogas plants** and promoting waste-to-energy technologies for sustainable waste disposal.

Read more: [Solid Waste Management \(SWM\) Rules, 2026](#)

Kerala's Oil Spill Contingency Plan



Source: TH

Why in News?

Kerala has finalised a draft **Oil Spill Contingency Plan (OSCP)** to protect the state's **590 km coastline**, following recent maritime disasters off the Kerala coast involving the **MSC Elsa 3** and **MV Wan Hai 503**.

- It has been developed under the **National Oil Spill Disaster Contingency Plan (NOS-DCP)** guidelines.

What are the Key Facts Regarding Kerala's Draft Oil Spill Contingency Plan?

- Environmental Sensitivity Mapping:** The plan includes **Environmental Sensitive Index (ESI)** mapping to identify and protect vulnerable marine ecosystems, including mangroves and fish breeding grounds.
- Shoreline Response & Management:** The OSCP establishes a clear chain of command for shoreline clean-up, incorporating **Net Environmental Benefit Analysis (NEBA)** and specific wildlife response protocols.
- Technological Integration:** It utilizes **hydrodynamic and oil spill modelling** to predict the trajectory of hazardous cargo, such as the **calcium carbide** and **plastic nurdles** lost during the MSC Elsa 3 sinking.
- Jurisdiction & Scope:** The plan covers marine spills within **12 nautical miles (Territorial Waters)** and extends **40 km** into inland riverine systems affected by tidal action.

What is the National Oil Spill Disaster Contingency Plan?

- About:** NOS-DCP is India's **apex** policy framework for responding to **oil and chemical spills** in Indian waters. Originally formulated in **1993** and promulgated in **1996**, the plan was significantly **updated in 2024** to address growing maritime traffic, larger tankers, offshore exploration, and emerging risks from **offshore wind energy**.
- Core Objectives:** Establish seamless coordination among various **ministries, ports, oil companies**, and **coastal states**.
 - Identify and protect ecologically sensitive zones (like **mangroves** and **coral reefs**) that require immediate protection during a spill.
 - Ensure that the **"Polluter Pays" principle** is enforced through legal and financial mechanisms.
- Key Features of NOS-DCP (2024 Updates):**
 - Integrated Digital Tracking:** Use of real-time satellite imagery, **Automatic Identification System (AIS)**, and the Online Oil Spill Advisory (OOSA) system for tracking and trajectory modelling of oil

spills.

- **Tiered Response Refinement:** The 2024 guidelines clarified the specific "Trigger Points" for different spill sizes:
 - **Tier 1 (Small):** Handled by the individual **port or oil facility** (Up to 700 tonnes).
 - **Tier 2 (Medium):** Requires regional resources/ **District administration** (700 to 10,000 tonnes).
 - **Tier 3 (Large):** Becomes a National Disaster, handled by the **ICG** and National Authorities (Over 10,000 tonnes).
- **Enhanced Chemical Preparedness:** With the increase in chemical transport, there is a heightened focus on **Hazardous and Noxious Substances (HNS)** .
- **Dispersant Policy:** New chemical dispersants were approved in 2024, focusing on "**low-toxicity**" **variants** to minimize the impact on marine life compared to older chemical standards.
- **Inclusion of Offshore Wind Farms:** It includes specific response protocols for the burgeoning offshore wind energy sector in India's **Exclusive Economic Zone (EEZ)** .
- **Institutional Framework:**

Entity	Role under NOS-DCP
Indian Coast Guard (ICG)	Indian Coast Guard (ICG) serves as the Central Coordinating Authority and maintains the National Inventory of pollution response equipment.
Directorate General of Shipping	Handles legal issues , ship-related regulations, and salvage operations.
INCOIS (Hyderabad)	Provides " Oil Spill Trajectory " models to predict where the oil will drift based on wind and currents.
Ministry of Environment (MoEFCC)	Identifies and prioritizes " Environmentally Sensitive Areas " (ESAs) for protection.

- **Major Exercise:** The ICG conducted the **10 th National Level Pollution Response Exercise (NATPOLREX-X)** off the coast of **Chennai** in October 2025 to test the efficiency of **inter-agency coordination** between the ICG, coastal states, and international observers (from 32 countries).

Oil Spill

- **About:** An **oil spill** is the accidental or intentional release of **liquid petroleum hydrocarbons** (crude oil or refined products) into the environment, particularly **marine ecosystems**.



- **Environmental Impact:** When oil spills into water, it forms a thin layer called an **oil slick** that spreads rapidly across the surface. The consequences are severe:
 - **Wildlife Damage:** **Oil coats** the feathers of birds and the fur of mammals (like sea otters), destroying their ability to insulate and stay buoyant, often leading to **hypothermia** or drowning.
 - **Ecosystem Disruption:** Thick oil can coat the roots of **mangroves** or settle on **coral reefs**, cutting off oxygen and light, leading to the **death of entire coastal "nurseries."**
 - **Human Impact:** Spills contaminate **local fisheries**, destroy tourism, and can even affect human health through the inhalation of toxic vapors or the consumption of contaminated seafood.

Frequently Asked Questions (FAQs)

1. What is the geographical scope of Kerala's OSCP?

It covers marine oil spills within 12 nautical miles (24 km) of Kerala's coastline (Territorial Waters) and extends 40 km inland into riverine systems or till tidal effect is evident, whichever is greater.

2. How does the 'Trigger Point' system work in NOS-DCPs?

It categorizes spills into Tier 1 (under 700 tonnes) for ports, Tier 2 (up to 10,000 tonnes) for regional resources, and Tier 3 (above 10,000 tonnes) as national disasters.

3. What role does INCOIS play during a maritime oil spill?

INCOIS (Hyderabad) provides real-time Oil Spill Trajectory models that predict the movement of the slick based on ocean currents, wind speed, and sea state.

UPSC Civil Services Examination Previous Year Question (PYQ)

Q. With reference to the United Nations Convention on the Law of Sea, consider the following statements: (2022)

1. A coastal state has the right to establish the breadth of its territorial sea up to a limit not exceeding 12 nautical miles, measured from baseline determined in accordance with the convention.
2. Ships of all states, whether coastal or land-locked, enjoy the right of innocent passage through the territorial sea.
3. The Exclusive Economic Zone shall not extend beyond 200 nautical miles from the baseline from which the breadth of the territorial sea is measured.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (d)

India's 1st Indigenous Glide Weapon TARA



Source: PIB

The **Defence Research and Development Organisation (DRDO)** and the **Indian Air Force (IAF)** achieved a significant milestone in indigenous defence technology by successfully conducting the maiden flight-trial of the **Tactical Advanced Range Augmentation (TARA) weapon off the coast of Odisha.**

- **TARA:** It is a modular **range extension kit** and is India's **first indigenous glide weapon system** designed to **convert unguided warheads (gravity bombs) into precision-guided weapons** , significantly enhancing strike capabilities.
- **Strategic Objective:** The system enhances the **range, lethality, and strike accuracy of low-cost conventional weapons** , enabling them to strike targets over long distances with greater precision.
 - The TARA glide weapon system can be launched from multiple fighter aircraft in the IAF's inventory, increasing operational flexibility.
- **Cost-Effectiveness:** It is the first glide weapon to utilize **state-of-the-art low-cost systems** , making high-precision strikes more economically sustainable.
- **Developmental Agency:** The system was designed and developed by the **Research Centre Imarat**

(RCI), Hyderabad , in collaboration with other **DRDO laboratories** .

- **Atmanirbhar Bharat:** The project involved **Development cum Production Partners (DcPP)** and Indian private industries, which have already commenced production activity, bolstering the **domestic defence manufacturing ecosystem** .
- **Operational Significance:** This development marks a major step in advancing India's **strategic autonomy** and reducing dependence on **imported precision-guided munitions (PGMs)**.



Read more: [Boosting Indigenisation & Innovation in Defence](#)